

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Third Semester of B. Tech. Examination (EE)

April/May 2012

EE - 207 Network Analysis

Date: 01/05/2012, Tuesday

Time: 1.30 pm to 4.30 p.m

Maximum Marks: 70

Instructions:

1. Attempt all questions from both sections.
2. Use separate answer book for each sections.
3. Figures to the right indicate full marks.
4. Assume suitable additional data, if necessary

SECTION - I

Q-1 A. Explain in brief ideal voltage source and ideal current source.

4

B. Define "Tree" and Co-Tree".

3

Q-2 A. For the circuit shown in Fig.1, draw the oriented graph and write the

7

- (i) incidence matrix
- (ii) cut-set matrix and
- (iii) tie-set matrix

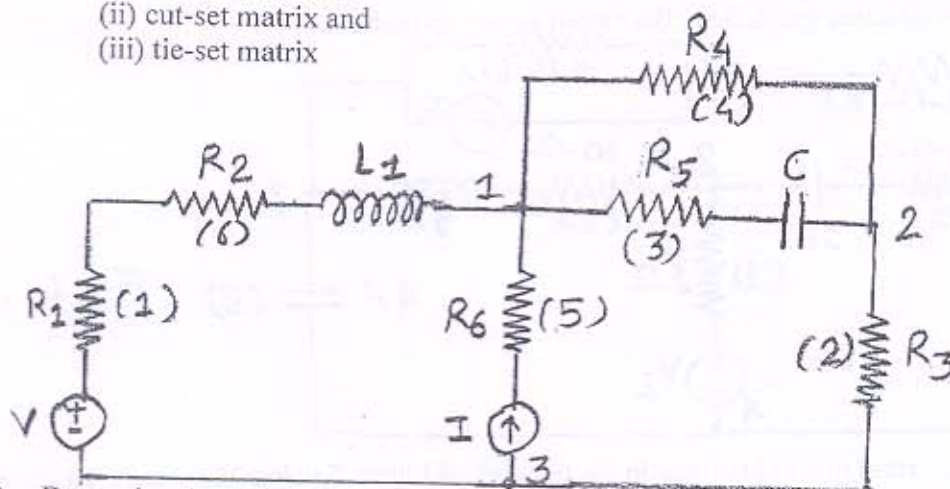


Fig. 1

B. Determine the voltage across 6 Ω resistor in the network of Figure 2 using nodal analysis.

5

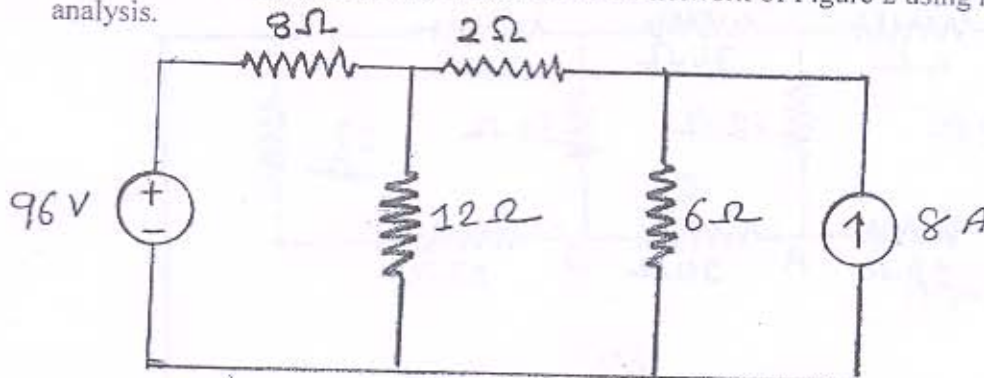


Fig. 2

C. What is Kirchhoff's current law (KCL)? Explain with example.

2

OR

- Q-2 A. Determine the current through $j10\ \Omega$ impedance in a network of Figure 3 using mesh current analysis. 7

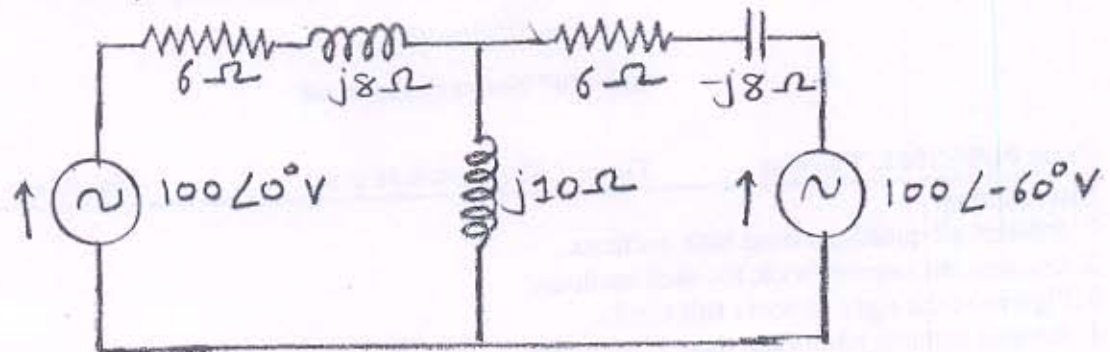


Fig. 3

- B. The fundamental cutset matrix of network is given as follows. 5

Twigs			Links		
a	c	e	b	d	f
1	0	0	1	0	1
0	1	0	0	-1	-1
0	0	1	1	1	1

Draw the oriented graph

- C. Draw the oriented graph from the circuit shown in figure 4. 2

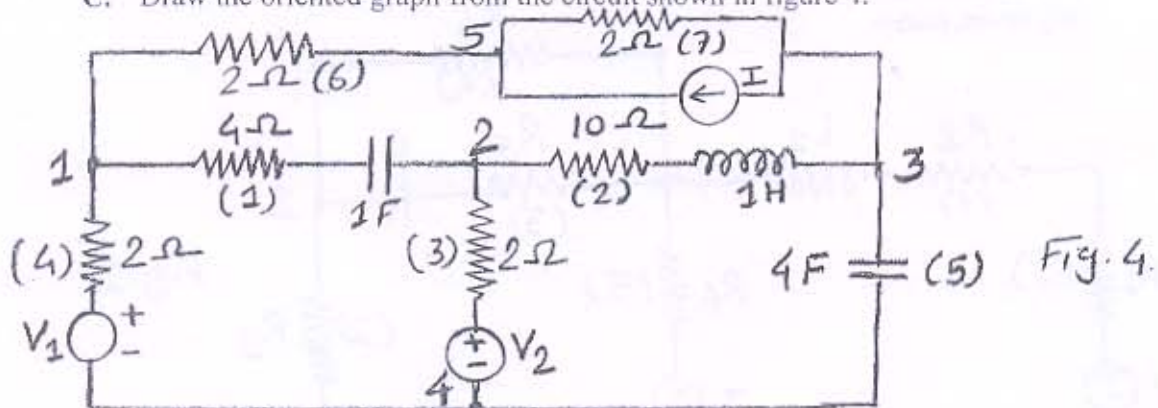


Fig. 4.

- Q-3 A. Find the current in $10\ \Omega$ resistor in the network of Figure 5 using Thevenin's theorem. Resistances are in ohms. 7

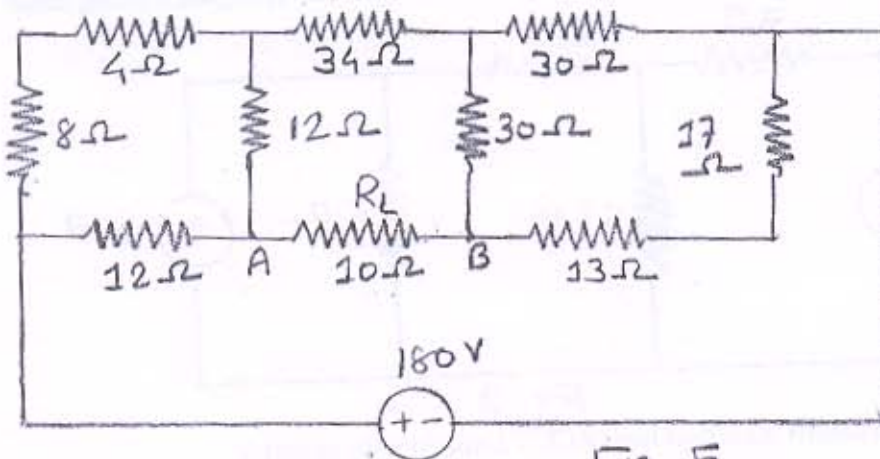


Fig. 5

- B. In the circuit of figure 6 the resistance R_S is variable between 2 and 50 ohms. What value of R_S results in maximum power transfer across the terminal AB. Also calculate the maximum power delivered to the load. 4

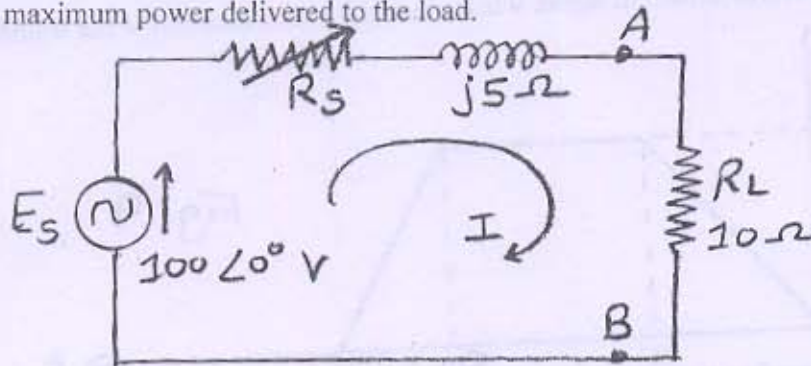


Fig. 6

- C. Define short-circuit admittance parameters with equation. 3

OR

- Q-3 A. Obtain z-parameters for the network shown in figure 7. 7

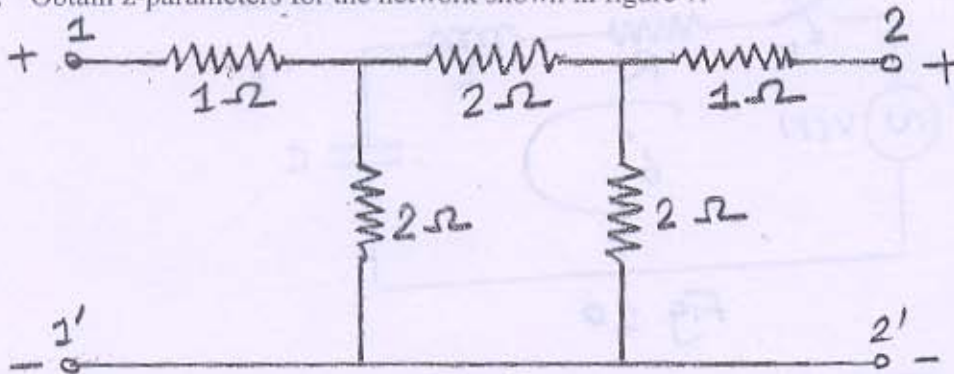


Fig. 7

- B. For the network shown in figure 8 determine h-parameters. 4

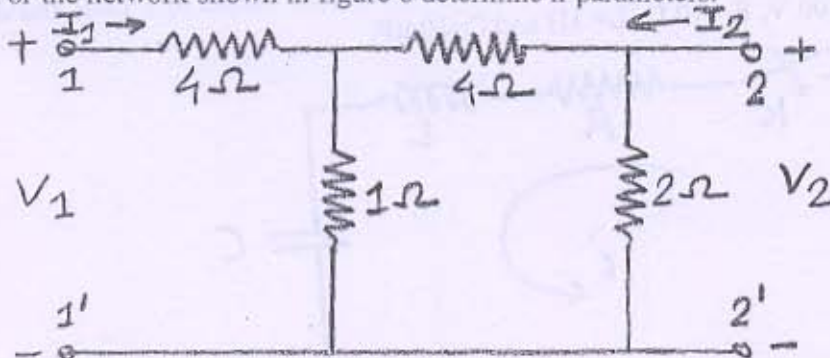
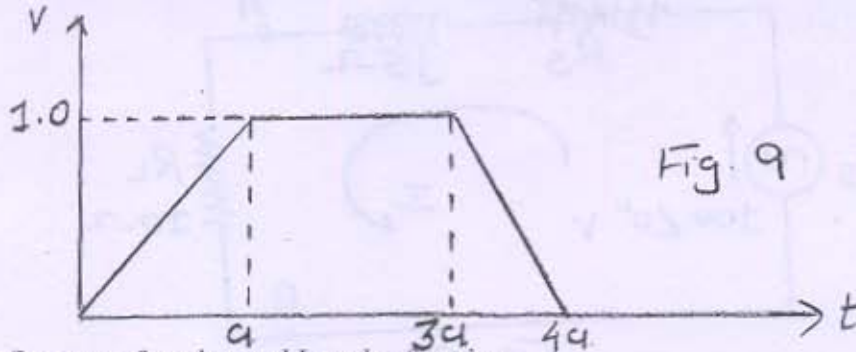


Fig. 8

- C. Explain in short reciprocity theorem. 3

SECTION - II

- Q-4 (A) The waveform shown in figure 9 is trapezoidal pulse. Determine the transform $V(s)$. 4



- (B) Define ramp function and Impulse function. 3

- Q-5 (A) In a series R-L-C circuit of figure 10, $R=5\ \Omega$, $L=1\text{ H}$, $C=0.25\text{ F}$ and $V(t)=6e^{-2t}$ volts. Switch k is closed at time $t=0$. Obtain particular solution for the current using Laplace transform method. Assume zero initial condition in the elements. 7

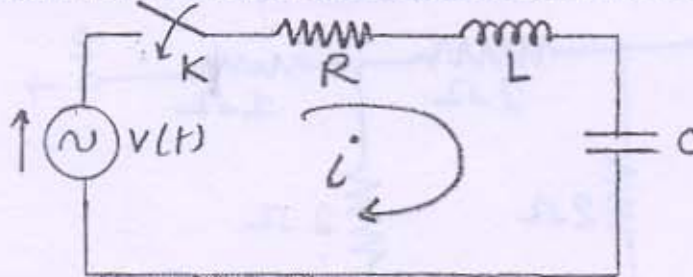


Fig. 10

- (B) In the network of the figure 11, the switch k is closed at $t=0$ with the capacitor uncharged and with zero current in the inductor. Find the values of i , di/dt and d^2i/dt^2 at $t=0^+$. If $v=100\text{ V}$, $R=10\ \Omega$, $L=1\text{ H}$ and $C=10\ \mu\text{F}$. 7

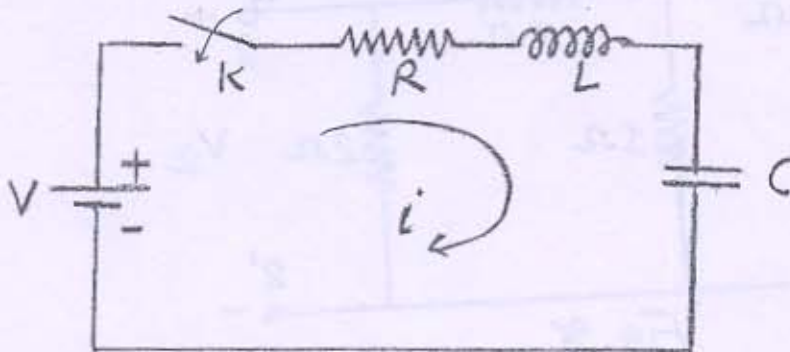


Fig. 11

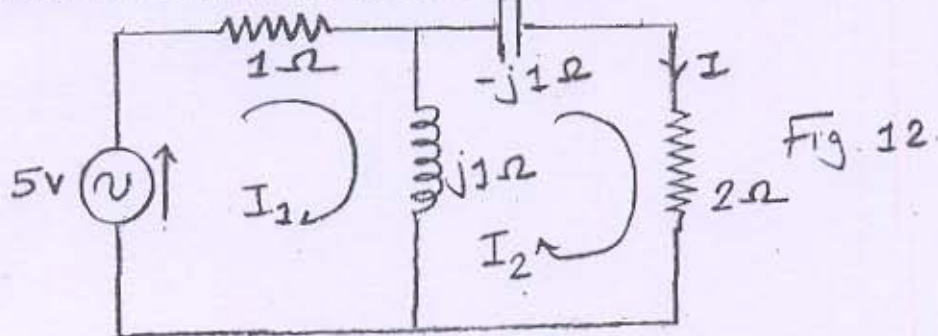
OR

- Q-5 (A) Explain in detail Initial and Final value theorem. 7
- (B) Explain the initial condition of Resistor, Inductor and Capacitor. 7
- Q-6 (A) The reduced incidence matrix of an oriented graph is given below. Draw the oriented 7

graph.

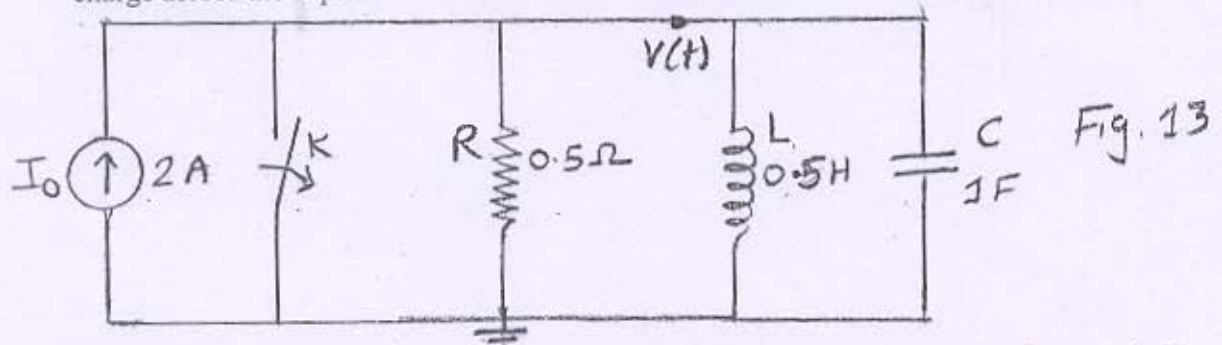
$$A = \begin{bmatrix} 0 & -1 & 1 & 1 & 0 \\ 0 & 0 & -1 & -1 & -1 \\ -1 & 0 & 0 & 0 & 1 \end{bmatrix}$$

- (B) In the network shown in figure 12, the voltage source of 5V causes a current I in the 2Ω resistor. Find I and verify the reciprocity theorem. 7



OR

- Q-6 (A) In the network of the figure 13, the switch k is opened at time $t=0$. Obtain the particular solution for voltage $v(t)$ across the parallel circuit. Assume that there is no charge across the capacitor C and no current in the inductor L before switching. 7



- (B) Determine the voltages of nodes 1 and 2 in the network of figure 14 using input and output admittances. 7

